

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo headphones / OPA1656 / relay mute

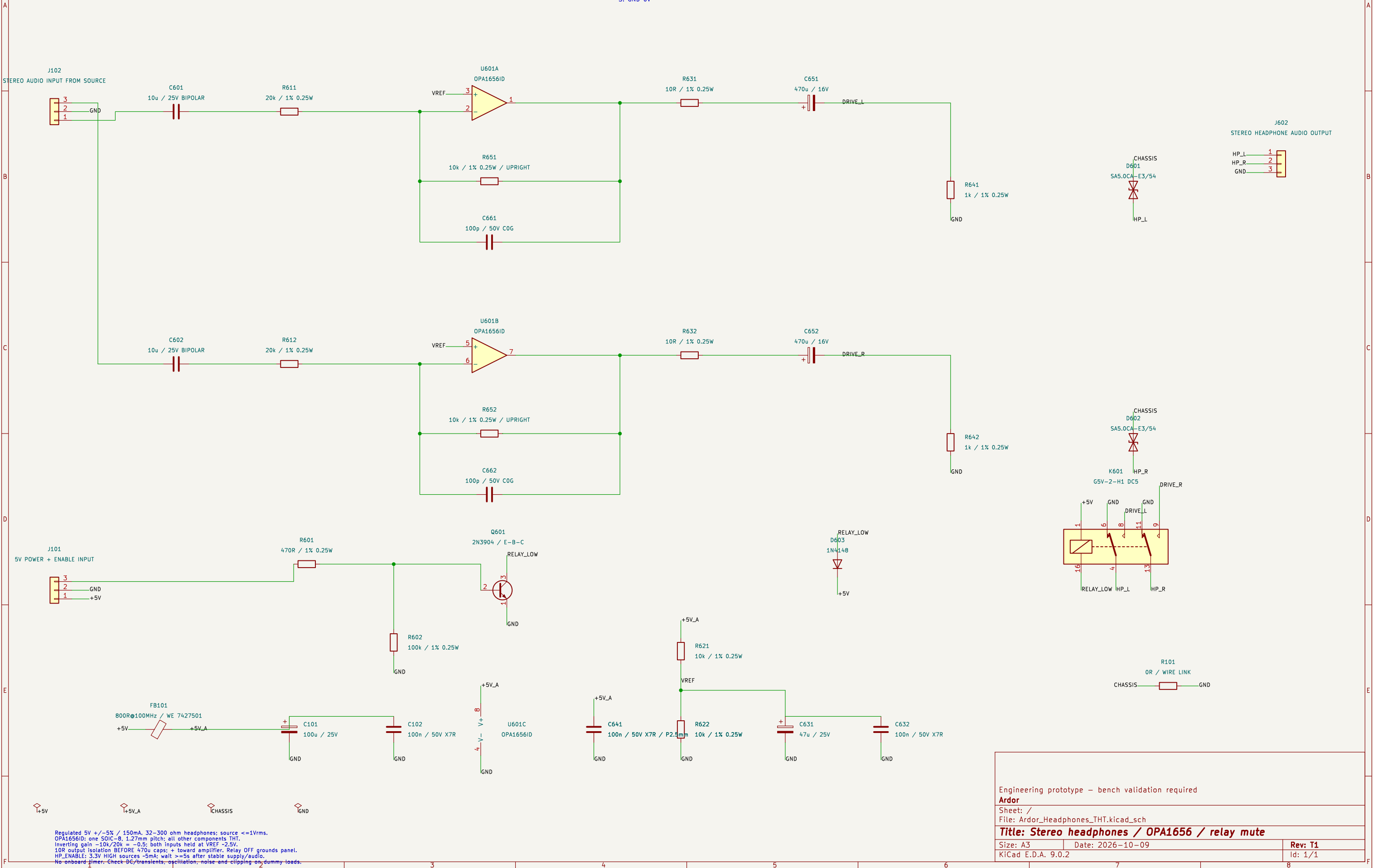
T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

J602 / HEADPHONE OUT
1: LEFT OUT 2: RIGHT OUT
3: GND 0V



Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Headphones_THT.kicad_sch

Title: Stereo headphones / OPA1656 / relay mute

Size: A3

Date: 2026-10-09

Rev: T1

KiCad E.D.A. 9.0.2

Id: 1/1

Regulated 5V +/-5% / 150mA. 32-300 ohm headphones; source <=1Vrms.
OPA1656ID: one 50IC-8, 1.27mm pitch; all other components THT.
Inverting gain -10k/20k = -0.5; both inputs held at VREF -2.5V.
10R output isolation BEFORE 470u caps; + toward amplifier. Relay OFF grounds panel.
HP_ENABLE: 3.3V HIGH sources -5mA; wait >=5s after stable supply/audio.
No onboard timer. Check DC/transients, oscillation, noise and clipping on dummy loads.